



Sri Lanka Institute of Information Technology

Project Management Tool Report

For

Oudra – A Smart Decision Support System for Sri Lankan Agarwood Plantations

Research Group ID: 25_26J_157

Submitted by:

Registration #	Student Name
IT22319142	WIJESINGHE A.G.T.
IT22123886	DEHIGOLLA D. N.
IT22123640	AMARASOORIYA R.P.C.A.
IT22310996	THEENNAKOON T.A.C.S.

Name of the supervisor

Ms. Chathurangika Kahandawaarachchi

Name of the co-supervisor

Ms. Narmada Gamage

Date of submission

12.04.2026

List of Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
CI/CD	Continuous Integration and Continuous Deployment
DB	Database
GIS	Geographic Information System
GPS	Global Positioning System
IDE	Integrated Development Environment
IEEE	Institute of Electrical and Electronics Engineers
IoT	Internet of Things
ISE	Information Systems Engineering
IT	Information Technology
LKR	Sri Lankan Rupee
ML	Machine Learning
CRUD	Create, Read, Update, Delete
NFC	Near Field Communication
PP1	Progress Presentation 1
QA	Quality Assurance
REST	Representational State Transfer
RP	Research Project
SDK	Software Development Kit

MQTT	Message Queuing Telemetry Transport
SRS	Software Requirement Specification
TAF	Topic Approval Form
UAT	User Acceptance Testing
UI/UX	User Interface User Experience
USD	United States Dollar

Table of Contents

List of Abbreviations	2
List of Figures.....	6
List of Tables.....	7
1. Executive Summary	8
1.1 Introduction	8
1.2 Objectives of the PM Tool Report	9
1.2.1 Main Objective.....	9
1.2.2 Specific Objectives	9
1.3 Report Scope.....	11
1.4 Project Team Overview	13
2. Project Management Tool Overview	15
2.1 Why we chose ClickUp	15
2.2 Communication & Collaboration Tools.....	16
2.3 Tool Integration Summary Table.....	17
3. Project Metrics & Task Summary	18
3.1 Task Allocation by Member	18
3.2 Task & Milestone Status View.....	20
3.3 Completed Tasks Breakdown	22
3.4 Current Week Progress Schedule.....	24
3.5 Dashboard View	25
3.6 Milestone Tracking Table	26
3.7 Priority Distribution Table	26
4. Resource & Work Management	27
4.1 Team Structure & Roles.....	27
4.2 Work Distribution.....	28
4.3 Risk Register.....	29

4.4 Budget	30
5. Communication Management	31
5.1 MS Teams Channel Structure	31
5.1.1 Shared General Folder Structure	32
5.1.2 Individual Member Folders	35
5.1.3 Meetings Folder	36
5.2 Meeting Log	36
6. Progress Summary	41
6.1 Achievements	41
6.2 Challenges Faced & Resolutions Table	43
6.3 Upcoming Tasks – Next Sprint	45
7. Research Panel Feedback	46
7.1 PP1 Feedback.....	46
7.2 PP2 Feedback.....	46
8. Lessons Learned	48
8.1 Successes	48
8.2 Challenges	48
8.3 Improvements for Future Projects.....	49
APPENDICES	50

List of Figures

Figure 3.1: Clickup Task Allocation 1.....	18
Figure 3.2: Clickup Task Allocation 2.....	18
Figure 3.3: Clickup Task Allocation 3.....	19
Figure 3.4: Clickup Task Allocation 4.....	19
Figure 3.5: Clickup Tasks & Milestone Statuses 1.....	20
Figure 3.6: Clickup Tasks & Milestone Statuses 2.....	20
Figure 3.7: Clickup Tasks & Milestone Statuses 3.....	21
Figure 3.8: Clickup Tasks & Milestone Statuses 4.....	21
Figure 3.9: Clickup Tasks & Milestone Statuses 5.....	22
Figure 3.10: Clickup Completed Tasks Breakdown 1.....	22
Figure 3.11: Clickup Completed Tasks Breakdown 2.....	23
Figure 3.12: Clickup Completed Tasks Breakdown 3.....	23
Figure 3.13: Clickup Completed Tasks Breakdown 4.....	24
Figure 3.14: Current Week Progress Schedule.....	24
Figure 3.15: Clickup Dashboard View 1.....	25
Figure 3.16: Clickup Dashboard View 2.....	25
Figure 3.17: Clickup Milestone Tracking Table.....	26
Figure 3.18: Clickup Priority Distribution Table.....	26
Figure 5.1: MS Teams Research Project Library Structure.....	31
Figure 5.2: MS Teams – General Folder Structure	34
Figure 5.3: MS Teams – Sample Individual Member Folder Structure	35

List of Tables

Table 1.1: Team Members and Individual Component Roles.....	13-14
Table 2.1: Tool Integration Summary.....	17
Table 4.1: Risk Register Table.....	29
Table 4.1: Budget in LKR.....	30
Table 5.1: MS Teams General Folder Content.....	32-33
Table 5.2: MS Teams – Individual Member Content.....	35
Table 5.3: Meeting Logs.....	36-40
Table 6.1: Challenges Faced and Resolutions.....	43-44
Table 7.1: Progress Presentation 1 Feedback.....	46
Table 7.2: Progress Presentation 2 Feedback.....	46-47

1. Executive Summary

1.1 Introduction

Agarwood (*Gyrinops walla*), a resinous heartwood species endemic to Sri Lanka, is one of the world's most commercially valuable natural products, contributing to a global market worth more than USD 32 billion. Despite its economic importance, agarwood crop management in Sri Lanka is highly reliant on manual, fragmented, and paper-based methods. Plantation managers have limited visibility into individual tree conditions, field operations require organized coordination, resin maturity is assessed visually without scientific support, and there is no credible traceability method across the agarwood supply chain. These deficiencies lead to premature harvesting, direct income loss, shorter tree lifespans, and lost possibilities in premium international markets.

The Oudra project was implemented to address these problems by creating an unified Decision Support System to facilitate commercial agarwood plantation management in Sri Lanka. The system combines four specialized components into a single, unified platform: a GPS-based Digital Tree Tracking and Virtual Tree Mapping system, an IoT Sensor Network for real-time environmental growth factor monitoring with ESP32 microcontrollers, an AI-assisted Resin Condition Assessment module powered by MobileNetV2, and a Blockchain-based Product Certification and Traceability Dashboard implemented on the Polygon Layer-2 network. These four components are linked together via a shared Node.js/Express RESTful backend, a MongoDB Atlas cloud database, a React web dashboard for managers and investors, and a React Native mobile application for field workers, resulting in a single source of truth spanning across all plantation activities.

This Project Management Tool Report formalizes the Oudra team's planning, coordination, and tracking methods throughout the project lifetime. The project team has utilized ClickUp as the primary project management tool for sprint planning, work allocation, milestone tracking, and backlog management. Microsoft Teams served as the key communication and collaboration platform, with organized channels for development, research, resources, and supervisor contacts. WhatsApp was used for quick day-to-day team coordination. GitHub offered version control with a systematic branching mechanism for all four component branches. VS Code, GitHub Desktop, Postman, Figma, and Microsoft Word were utilized for development, API testing, UI design, and documentation workflows, respectively while using canva for the presentation slide preparations.

This report outlines the entire team's project management methodology, including sprint planning across the development iterations, task allocation per member, milestone tracking against deadlines, risk identification and mitigation, communication management, supervisor feedback from progress presentation reviews (PP1, PP2), and lessons learned. It is a documented record of the team's systematic and methodical approach to delivering the Oudra system and was developed for submission as part of the final year research project requirements for the academic year 2025/2026.

1.2 Objectives of the PM Tool Report

1.2.1 Main Objective

The primary goal of this Project Management Tool Report is to provide an in-depth and transparent summary of the project management practices, tools, processes, and progress employed by the Oudra team during the development lifecycle of the Decision Support System for Agarwood Plantation Management, emphasizing the team's structured, methodical, and collaborative approach to planning, executing, and tracking a complicated four-component final-year research project.

1.2.2 Specific Objectives

The following specific objectives define what this report aims to document and demonstrate:

- To provide a reasoned overview of the project management and collaboration technologies chosen by the team, notably ClickUp, Microsoft Teams, GitHub, VS Code, Postman, and Figma, including their justification and integration into the team's workflow.
- To record the whole task allocation, sprint planning, and backlog management approach used during the development sprints, demonstrating systematic effort distribution among the four team members and their unique research components.
- To document and track all project milestones against planned and actual dates, including TAF submission, field visit, SRS submission, Progress Reviews (PP1, PP2), conference paper submission, and final system delivery, to demonstrate milestone adherence and timeline management.

- To identify, evaluate, and document project risks and mitigation measures, demonstrating the team's collaborative approach to addressing technical, operational, and academic challenges throughout the project.
- To formally document the team's communication system, including MS Teams channel structuring, meeting logs, and important technical decision notes, allowing for traceability of all major project decisions.
- To document supervisor input received during official progress reviews, as well as the actions taken to demonstrate the team's adherence to academic instructions.
- To summarize the project's lessons learned, including accomplishments, obstacles encountered and overcome, and improvements identified, in order to contribute to the team's reflective practice and serve as a reference for future research projects.

1.3 Report Scope

This Project Management Tool Report presents the entire project management lifecycle of the Oudra research project, which was carried out by a team of four BSc (Hons) Information Technology - Information Systems Engineering students at the Sri Lanka Institute of Information Technology (SLIIT) during the academic year 2025/2026. The report covers all planning, tracking, coordination, communication, and review activities from the project's inception to the current level of development, and it provides a formal framework for tracking continuing progress until the final submission.

Within the scope of this report:

- **Project Management Tool Usage** - The report details the team's complete use of ClickUp for sprint planning, task assignment, backlog management, and progress monitoring during all the development sprints. It outlines how tasks were created, prioritized, assigned to specific individuals, and tracked until completion within the ClickUp workspace.
- **Task and Workload Management** - The scope includes task allocation documentation for all four individual research components (Digital Tree Tracking, IoT Sensor Monitoring, AI Resin Detection, and Blockchain Certification), as well as shared team responsibilities such as backend development, system integration, testing, and journal, conference paper submissions.
- **Sprint and Milestone Planning** - The report covers all project sprints, from Sprint 1 (Literature Review and TAF Submission, June 2025) to Sprint 8 (Full QA Testing and Final Submission), and includes planned versus actual milestone dates for all key deliverables such as the SRS submission, Progress Reviews PP1, PP2, Elsevier journal paper submission and conference paper submission to conferences such as IEEE TENSYP 2026, ICECET 2026.
- **Risk Management** - The identification, assessment, and mitigation of project risks unique to the Oudra system are documented, including technical risks related to IoT hardware, blockchain testnet availability, AI model performance, GPS reliability, and academic challenges such as conference paper revision cycles.
- **Communication and Collaboration** - The scope includes the team's systematic use of Microsoft Teams channels, WhatsApp for daily coordination, and GitHub for version-controlled collaboration. It comprises the meeting logs for every sprint planning meeting, supervisor

progress reviews, and field visit debriefs, as well as the important technical decision record that was kept throughout the project.

- Supervisor Feedback and Action Items - This report includes formal feedback from Progress Reviews PP1 and PP2, as well as the relevant action items raised and how the team addressed them.
- Budget and Resource Management - The report includes a financial summary for the project in Sri Lankan rupees (LKR), broken down into equipment, publication, tools and technology, and data gathering expenses.
- Lessons Learned - The scope includes a systematic reflection on the team's accomplishments, obstacles encountered and overcome, and recommendations for future initiatives.

Outside of the scope of this report:

- Technical implementation details, system architecture, and software design standards are addressed in the separately submitted Software Requirements Specification (SRS) document (Group ID: 25-26J-157).
- The conference paper submitted to the conferences such as IEEE TENSYPMP 2026, IEEE ICECET 2026 and the Elsevier journal papers document the Oudra system's research findings, literature review, methodology, and academic contributions.
- The final system submission includes component-level technical documentation such as API specifications, database schemas, model training details, and smart contract code.

1.4 Project Team Overview

The Oudra research project was designed by a group of four final-year BSc (Hons) IT - Information Systems Engineering students at the Sri Lanka Institute of Information Technology. Each team member is fully responsible for one independently scoped research component while also contributing to the shared system infrastructure, backend integration, and IEEE conference paper. The team is guided by a designated research supervisor through planned Progress Reviews and continuing interactions via Microsoft Teams.

The table below summarizes each team member's specific research component, key technology assignments, and current contribution status.

Table 1.1: Team Members and Individual Component Roles

Student ID	Name	Individual Component	Role and Responsibilities	Primary Tech Stack	Current Status
IT22319142	WIJESINGHE A.G.T.	Digital tree tracking and virtual tree mapping system	GPS coordinate capture, NFC card scanning for tree identification, real time virtual map rendering, React Native mobile app development, field note logging, router optimization for daily tasks, web app development	React Native, Node.js, Express, React, MongoDB Atlas, Google Maps API, NFC, Directions API, Maps SDK For Android, Distance Matrix API, Geolocation API, Maps JavaScript API, Maps SDK for Android, Routes API	Completed , On Track
IT22123886	DEHIGOLLA D.N.	IoT Sensor Network & Growth Factor Analysis System	ESP32 microcontroller programming, MQTT data pipeline, environmental	ESP32, MQTT, Node.js, MongoDB Atlas, React Native,	Completed , On Track

			parameter monitoring (pH, temperature, soil moisture, humidity), threshold-based auto task generation	Node.js, Express, React, Weather API	
IT22123640	AMARASOORIYA R.P.C.A.	AI-Assisted Resin Condition Assessment	MobileNetV2 model training and optimization, plantation image dataset collection and preprocessing, mobile camera integration, resin classification workflow	Python, TensorFlow, MobileNetV2, React Native, Node.js, Express, React, MongoDB Atlas	Completed On Track
IT22310996	THENNAKOON T.A.C.S.	Blockchain Certification & Traceability	Solidity smart contract development, Polygon Amoy testnet deployment, traceability dashboard on React web app, blockchain initialization and transaction management	Solidity, Polygon Layer-2, Web3.js, React, MongoDB Atlas, Express, Node.js	Completed On Track

2. Project Management Tool Overview

2.1 Why we chose ClickUp

ClickUp was selected as the primary project management tool due to its ability to handle agile project management principles, task coordination, and real-time progress tracking all on a single unified platform. Given the multi-component nature of our research project, which included IoT integration, AI model building, blockchain implementation, and web/mobile application development, we needed to use a solution that allowed for centralized planning and structured work management.

ClickUp's advanced capabilities, including task creation, task dependencies, priority settings, milestone tracking, sprint planning, and customized workflows, enabled the team to successfully manage complicated development tasks. Each system component was divided into small tasks and allocated to team members with specific timeframes, assuring clear duty and accountability.

Furthermore, ClickUp provides real-time progress tracking through dashboards and reporting features, allowing teams to analyze task completion status and detect bottlenecks early on. The program also helped to manage both technical development chores and research-related documents, such as SRS preparation, proposal writing, and report development. ClickUp's ability to integrate with other tools and maintain a structured workflow makes it ideal for directing the growth of a multidisciplinary research project, which ultimately improves team productivity, communication, and project visibility.

2.2 Communication & Collaboration Tools

Effective communication and collaboration mechanisms were critical for ensuring smooth coordination among team members throughout the project lifecycle. A combination of communication and collaboration tools was utilized to support real-time interaction, document sharing, and version control.

- **WhatsApp** was primarily used for informal and instant communication among team members. It enabled quick discussions, task reminders, and immediate updates regarding development progress.
- **Microsoft Teams and Zoom** were used for formal meetings, including discussions with supervisors, sprint reviews, and decision-making sessions. These platforms facilitated structured communication and allowed the team to clarify technical and research-related aspects effectively.
- **GitHub** (Web and Desktop) served as the primary collaboration platform for source code management. It enabled multiple team members to work on different components simultaneously while maintaining version control. Features such as branching, merging, and commit tracking ensured that all code changes were properly managed and documented.
- **Google Drive / OneDrive** were used for storing and sharing project-related documents, including the SRS, proposal, reports, and presentation materials. This ensured that all team members had access to the latest versions of documents at any time.

Overall, the integration of these tools enabled efficient communication, seamless collaboration, and proper coordination between development and research activities, which contributed significantly to the successful progress of the project.

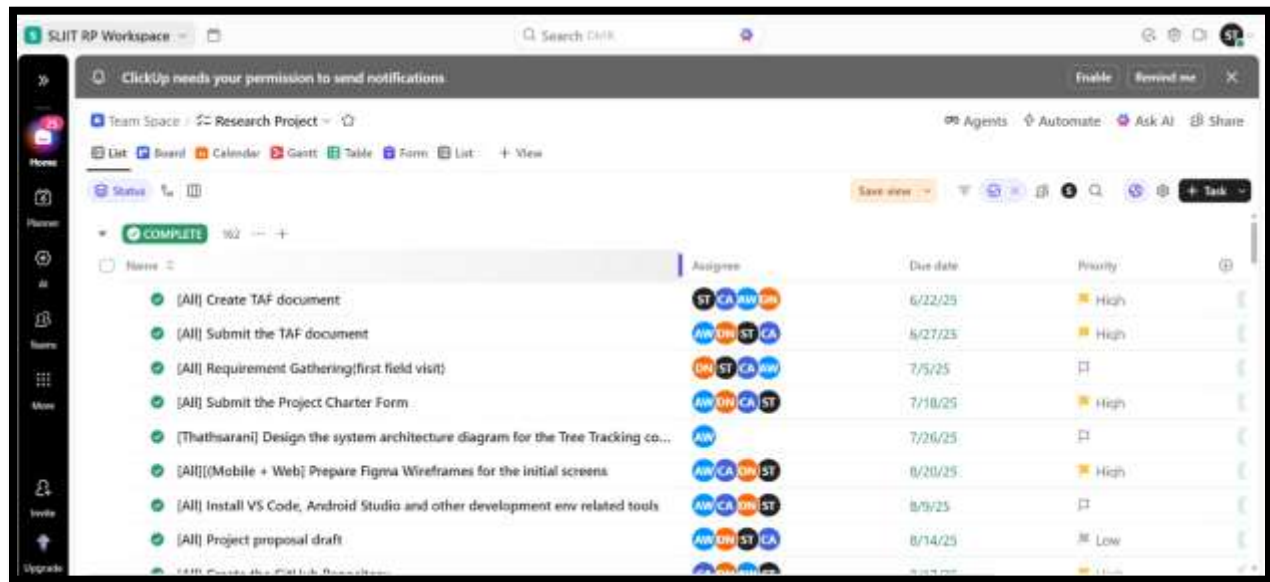
2.3 Tool Integration Summary Table

Table 2.1: Tool Integration Summary

Tool	Category	Purpose	Usage in Project
ClickUp	Project Management	Task management and sprint planning	Task allocation, progress tracking, milestone management
Visual Studio Code	Development	Code development environment	Frontend (React), Backend (Node.js) development
GitHub Desktop	Version Control	Local repository management	Commit changes and synchronize with remote repository
GitHub Web	Collaboration Platform	Code hosting and version tracking	Repository management, collaboration, issue tracking
Postman	API Testing	Backend API testing and validation	Testing REST APIs and verifying responses
Microsoft Word	Documentation	Report and document preparation	SRS, proposal, PM report, and final documentation
WhatsApp	Communication	Instant messaging	Quick updates and coordination
Zoom / MS Teams	Communication	Online meetings	Supervisor discussions and team meetings
Google Drive / OneDrive]	File Sharing	Document storage and sharing	Sharing reports and research materials

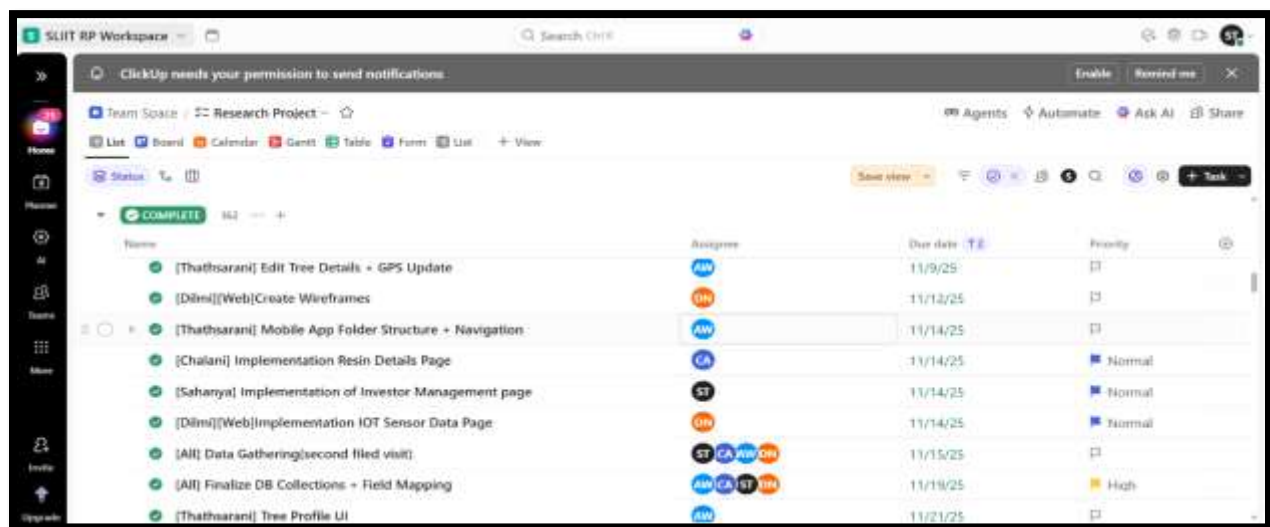
3. Project Metrics & Task Summary

3.1 Task Allocation by Member



Name	Assignee	Due date	Priority
[All] Create TAF document	ST CA AW DH	6/22/25	High
[All] Submit the TAF document	AW DH ST CA	6/27/25	High
[All] Requirement Gathering(first field visit)	DH ST CA AW	7/5/25	Medium
[All] Submit the Project Charter Form	AW DH CA ST	7/18/25	High
[Thatthasrani] Design the system architecture diagram for the Tree Tracking co...	AW	7/26/25	Medium
[All]((Mobile + Web) Prepare Figma Wireframes for the initial screens	AW CA DH ST	8/20/25	High
[All] Install VS Code, Android Studio and other development env related tools	AW CA DH ST	8/9/25	Medium
[All] Project proposal draft	AW DH ST CA	8/14/25	Low

Figure 3.1: Clickup Task Allocation 1



Name	Assignee	Due date	Priority
[Thatthasrani] Edit Tree Details + GPS Update	AW	11/9/25	Medium
[Dilini][Web]Create Wireframes	DH	11/12/25	Medium
[Thatthasrani] Mobile App Folder Structure + Navigation	AW	11/14/25	Medium
[Chalani] Implementation Resin Details Page	CA	11/14/25	Normal
[Sahanya] Implementation of Investor Management page	ST	11/14/25	Normal
[Dilini][Web]Implementation IOT Sensor Data Page	DH	11/14/25	Normal
[All] Data Gathering(second filed visit)	ST CA AW DH	11/15/25	Medium
[All] Finalize DB Collections + Field Mapping	AW CA ST DH	11/19/25	High
[Thatthasrani] Tree Profile UI	AW	11/21/25	Medium

Figure 3.2: Clickup Task Allocation 2

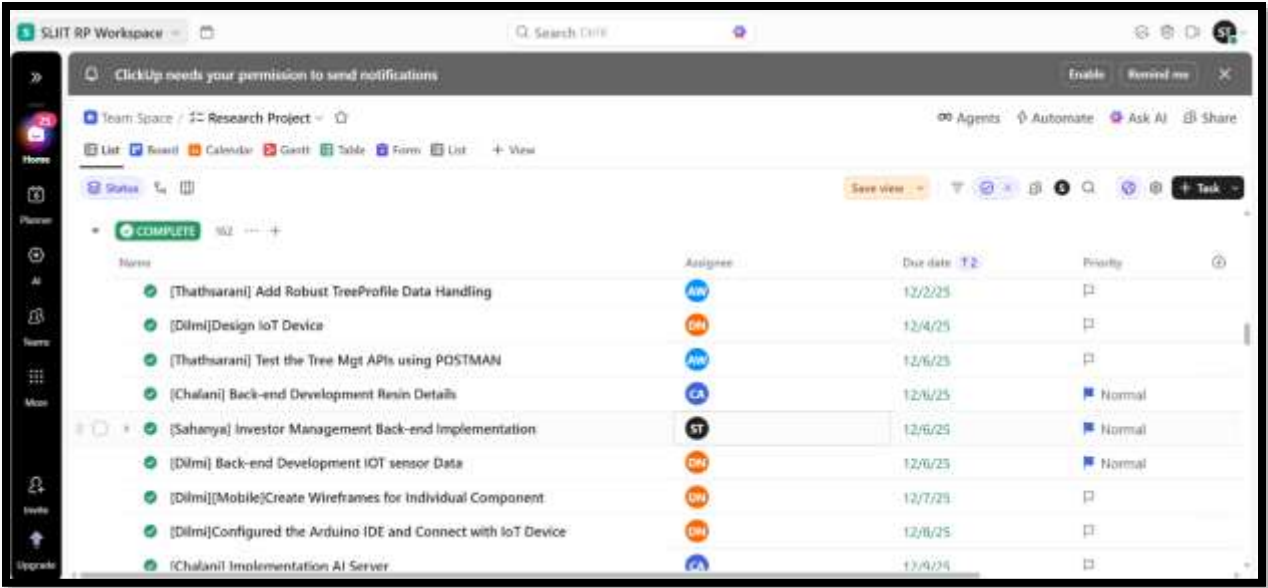


Figure 3.3: Clickup Task Allocation 3

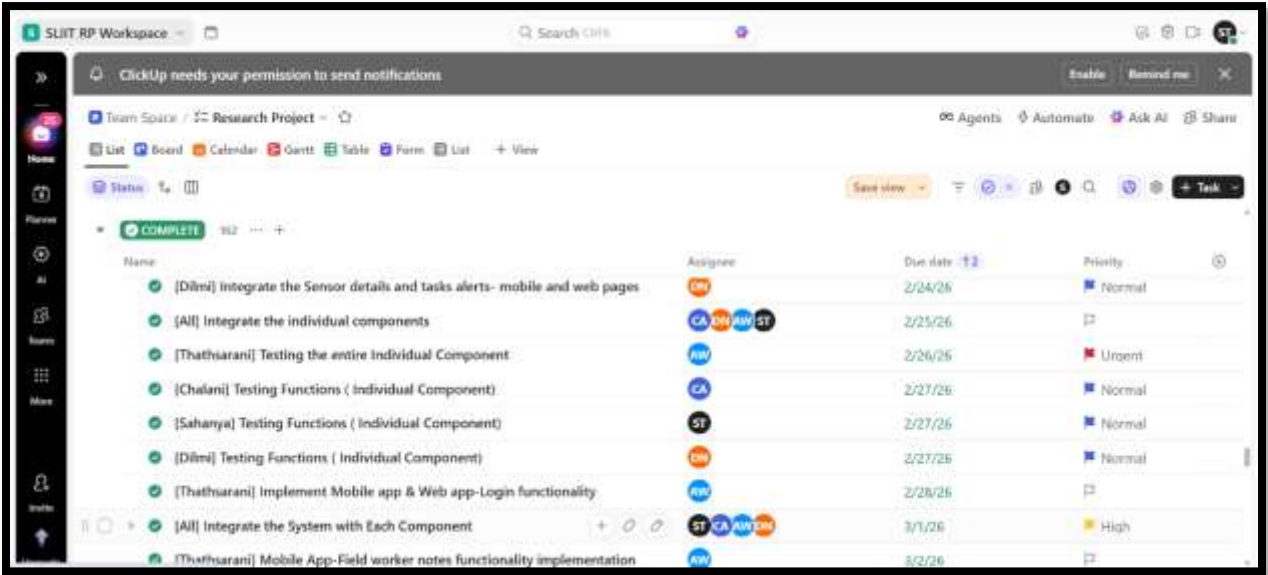


Figure 3.4: Clickup Task Allocation 4

3.2 Task & Milestone Status View

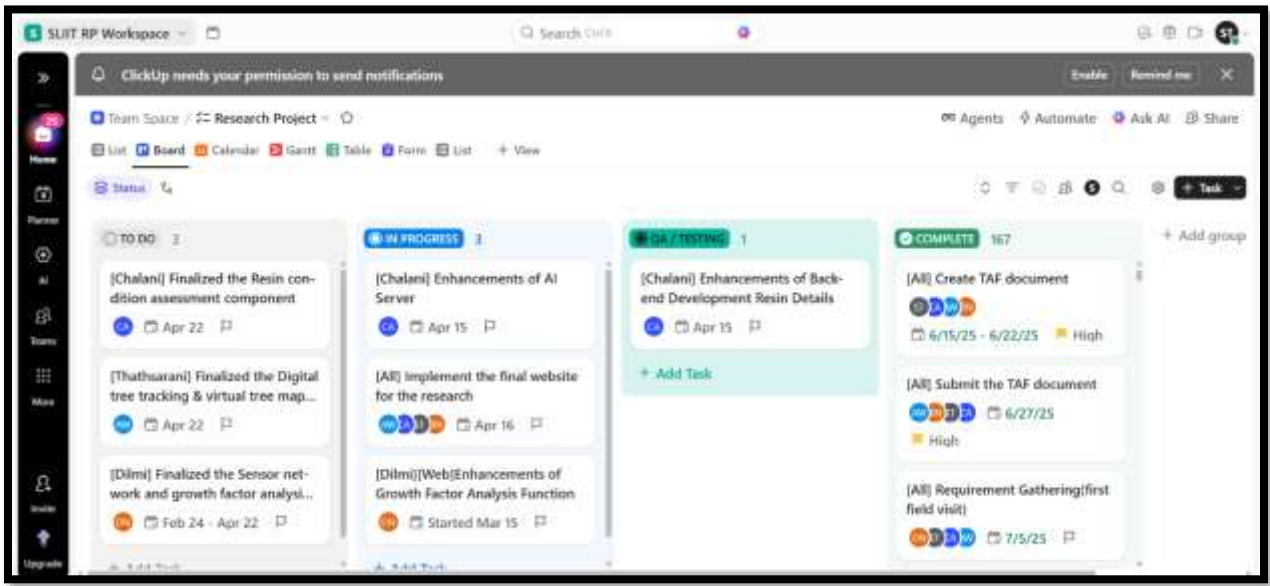


Figure 3.5: Clickup Tasks & Milestone Statues 1

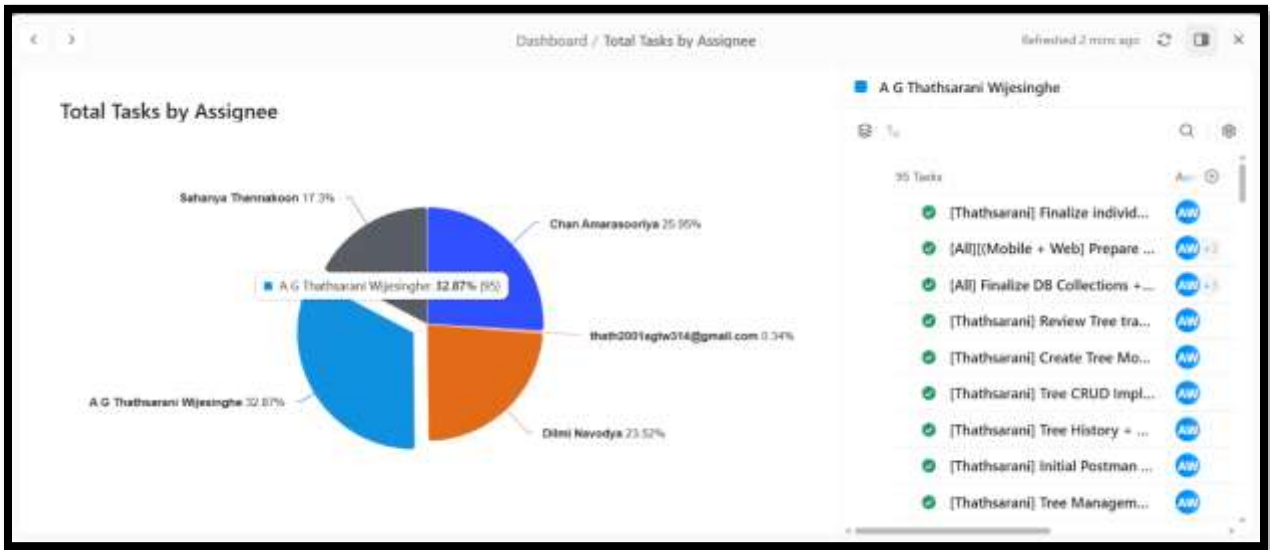


Figure 3.6: Clickup Tasks & Milestone Statues 2

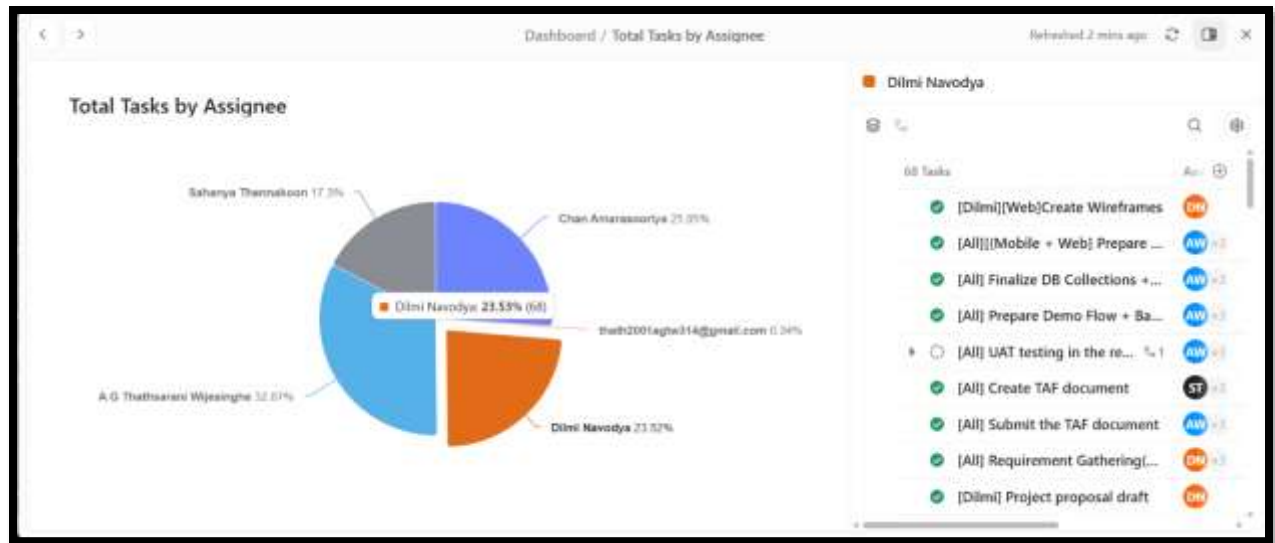


Figure 3.7: Clickup Tasks & Milestone Statuses 3

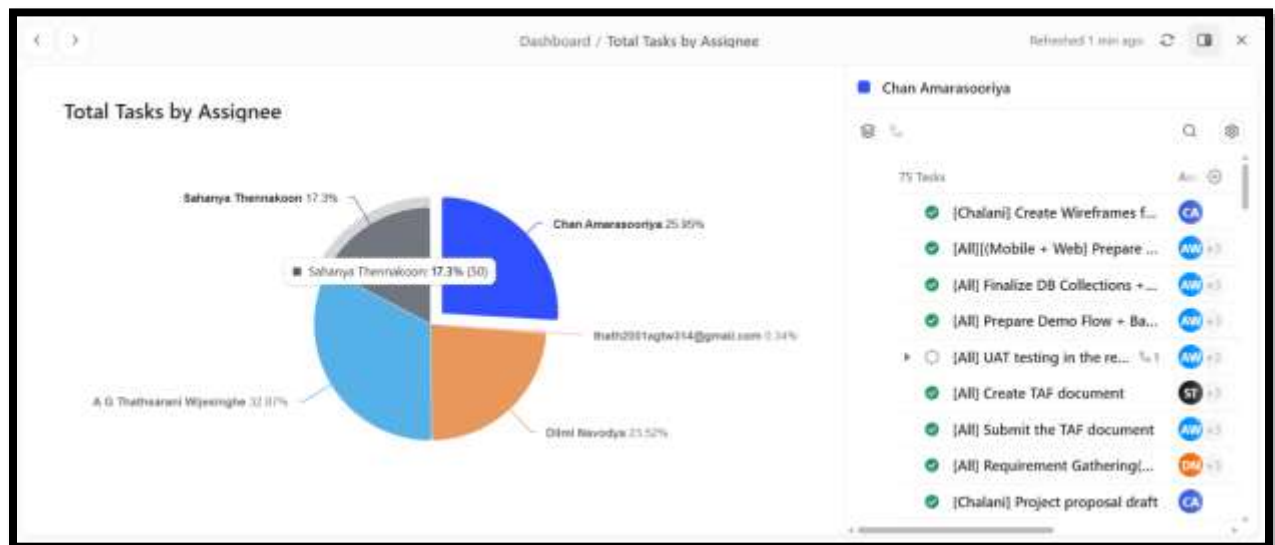


Figure 3.8: Clickup Tasks & Milestone Statuses 4

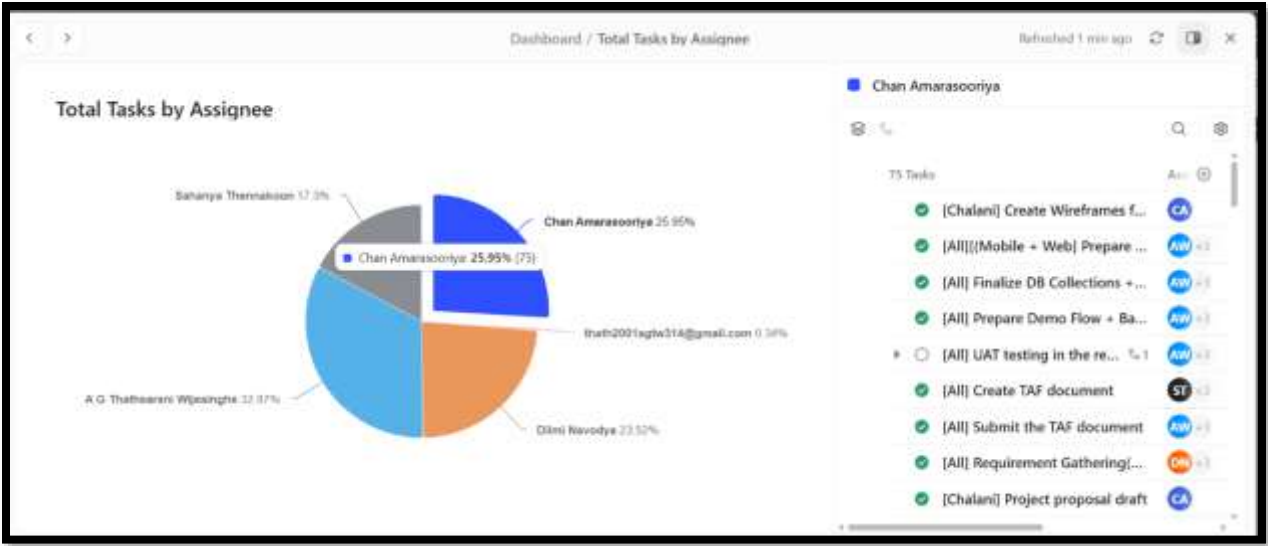


Figure 3.9: Clickup Tasks & Milestone Statuses 5

3.3 Completed Tasks Breakdown

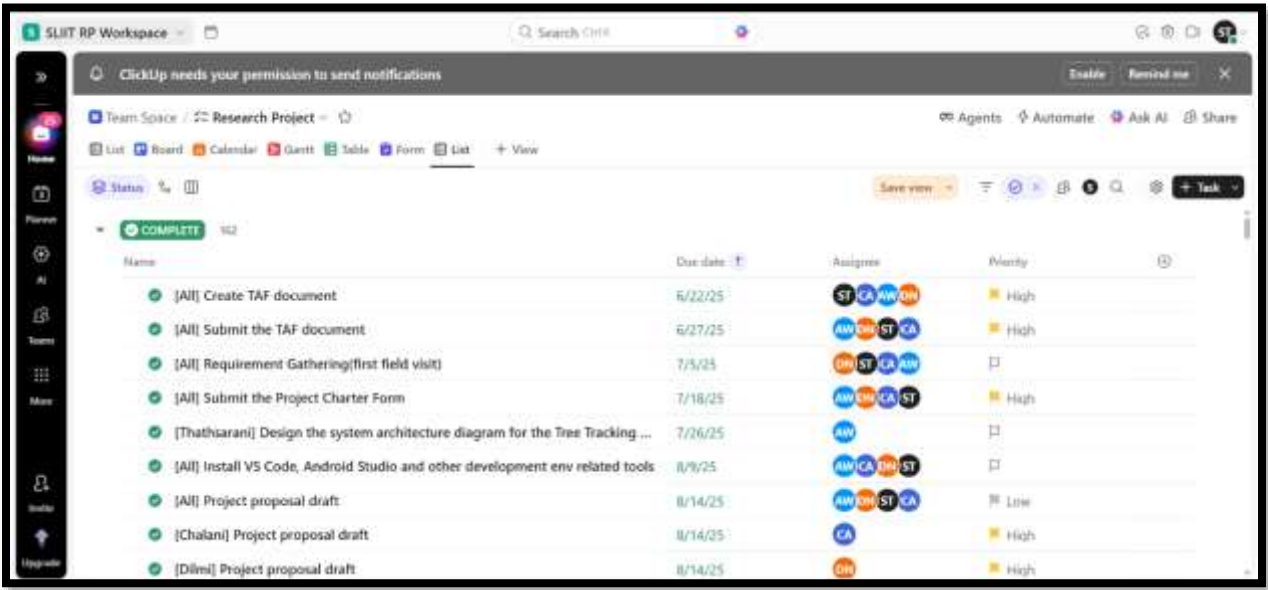


Figure 3.10: Clickup Completed Tasks Breakdown 1

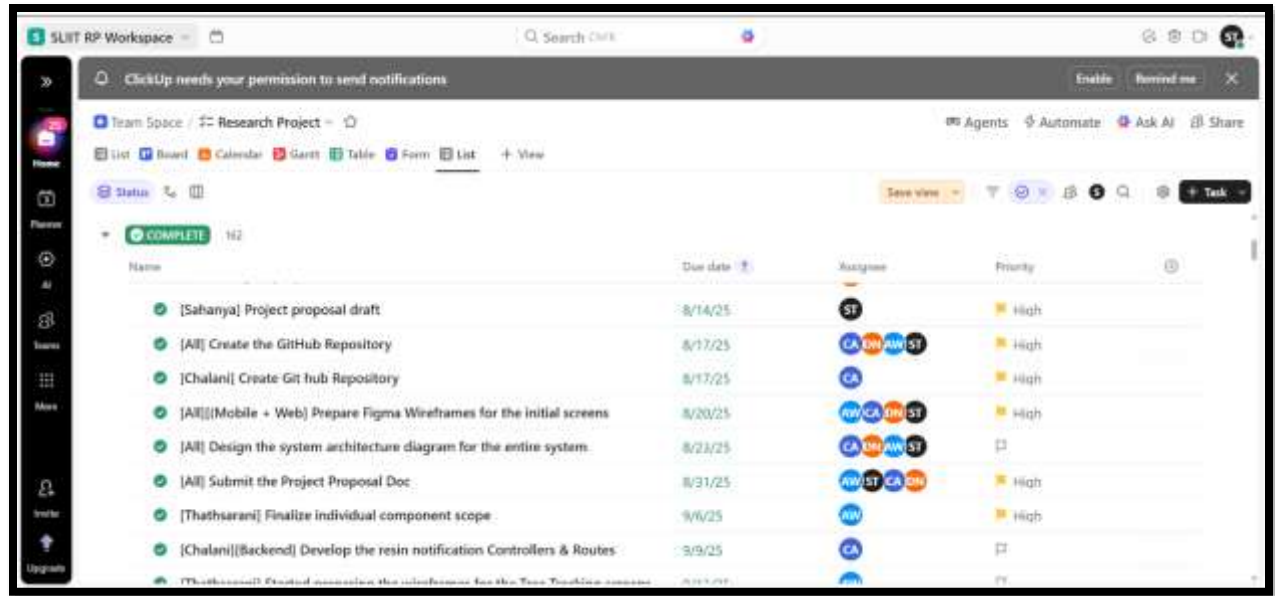


Figure 3.11: Clickup Completed Tasks Breakdown 2

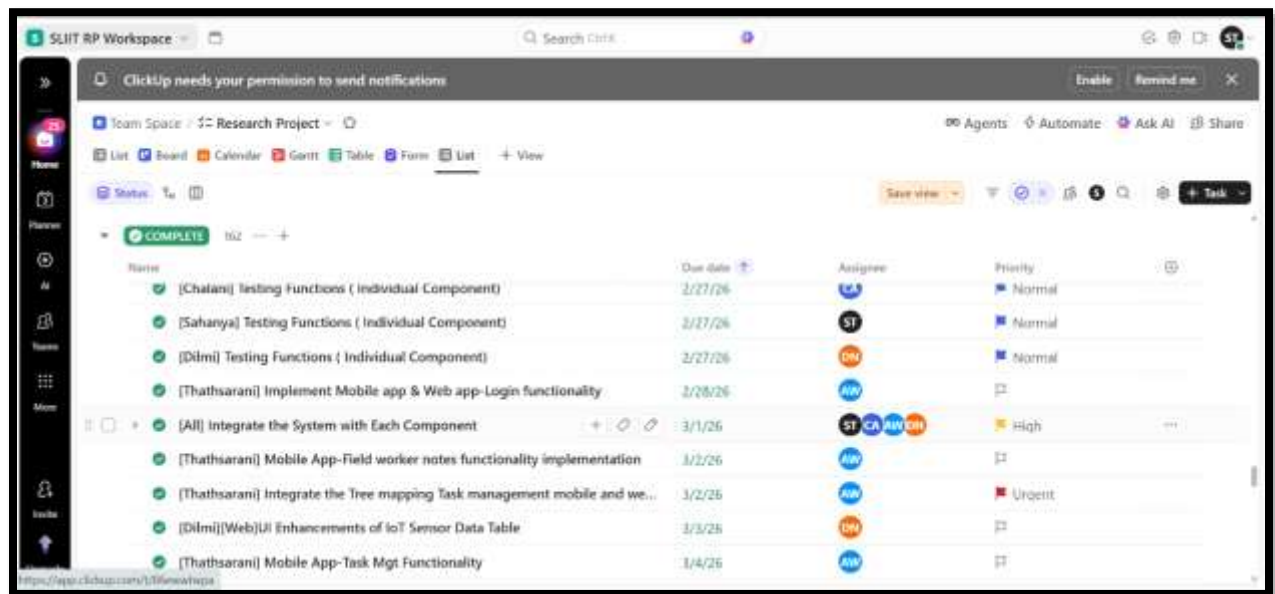


Figure 3.12: Clickup Completed Tasks Breakdown 3

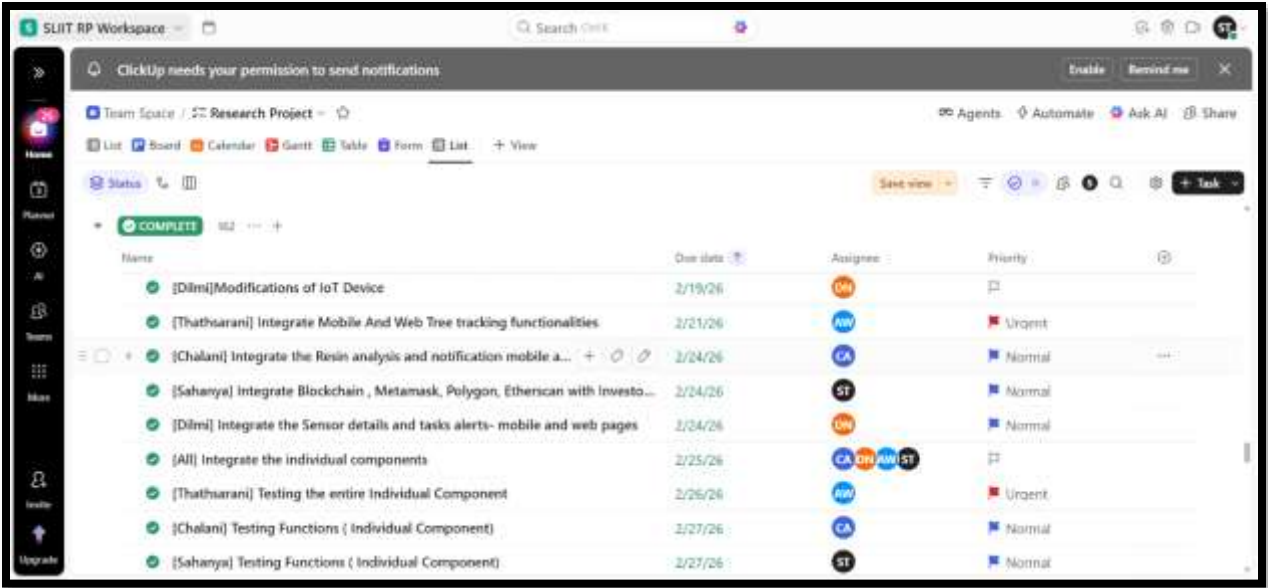


Figure 3.13: Clickup Completed Tasks Breakdown 4

3.4 Current Week Progress Schedule

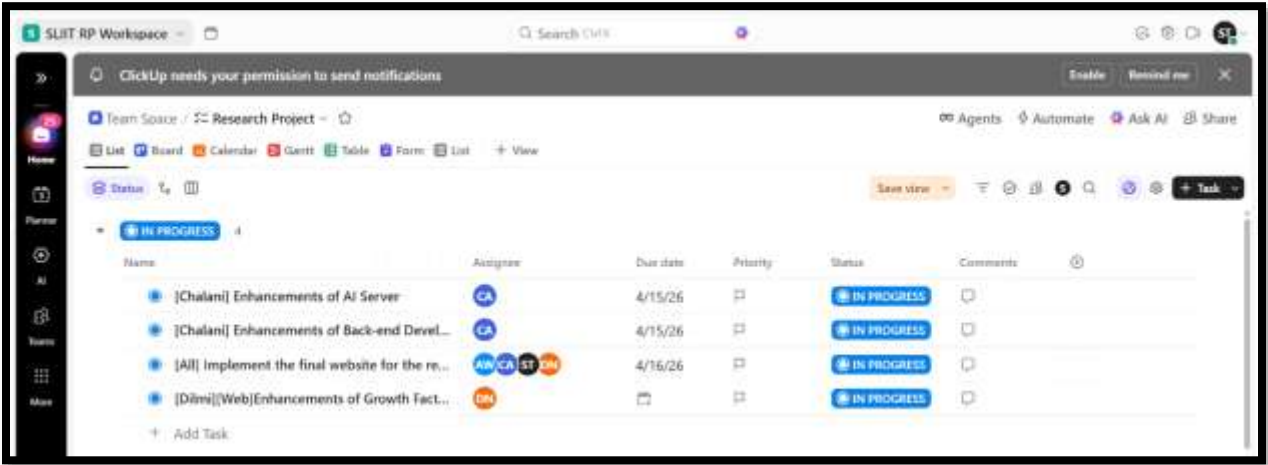


Figure 3.14: Current Week Progress Schedule

3.5 Dashboard View

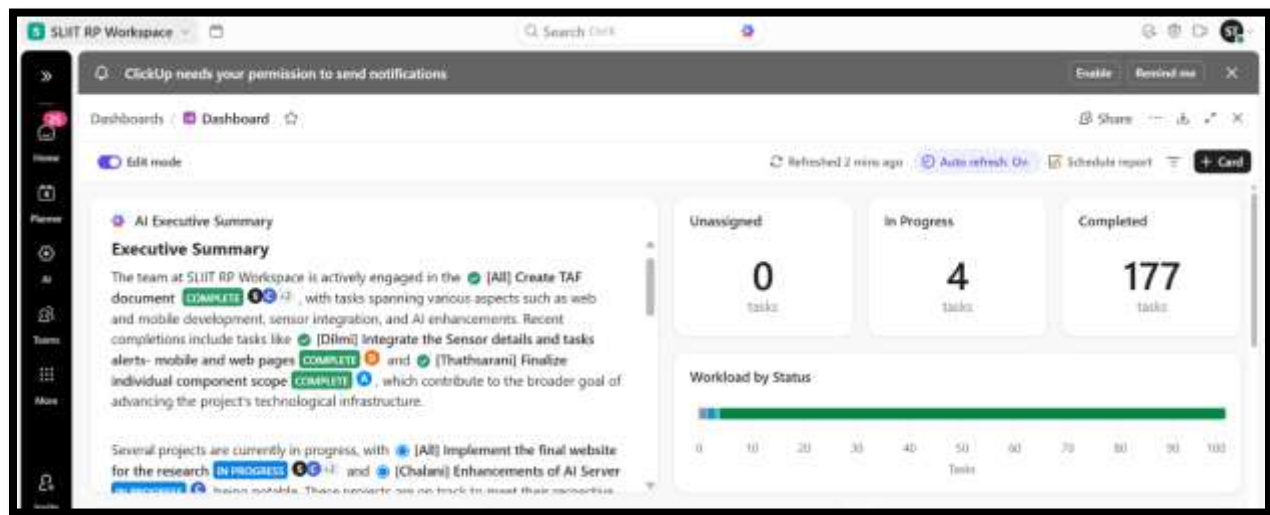


Figure 3.15: Clickup Dashboard View 1

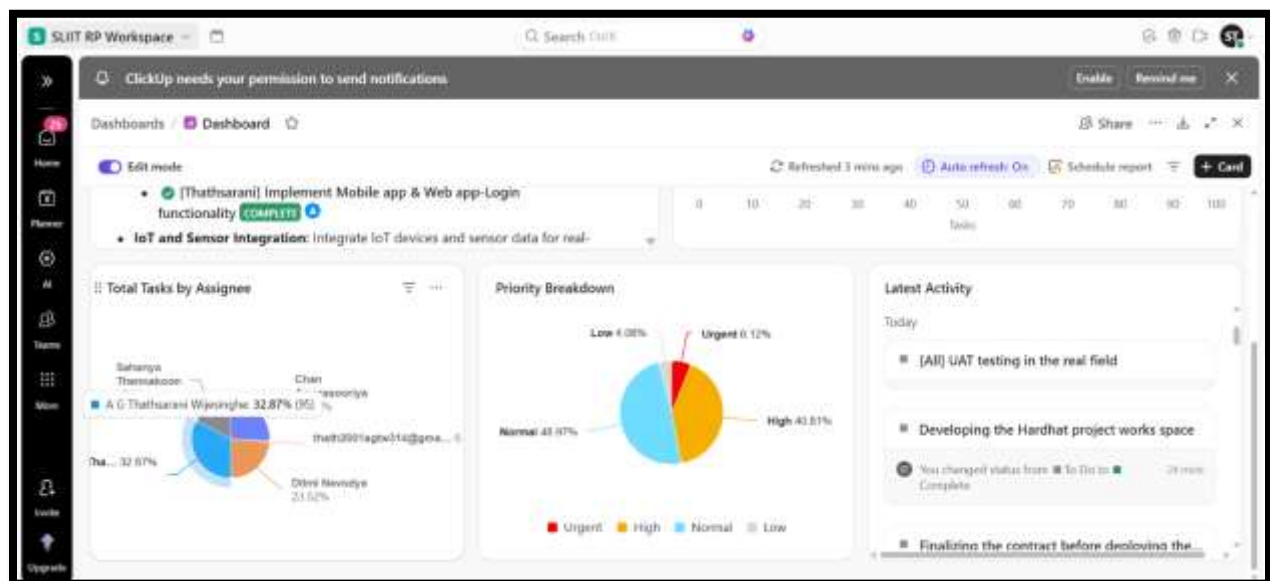


Figure 3.16: Clickup Dashboard View 2

3.6 Milestone Tracking Table

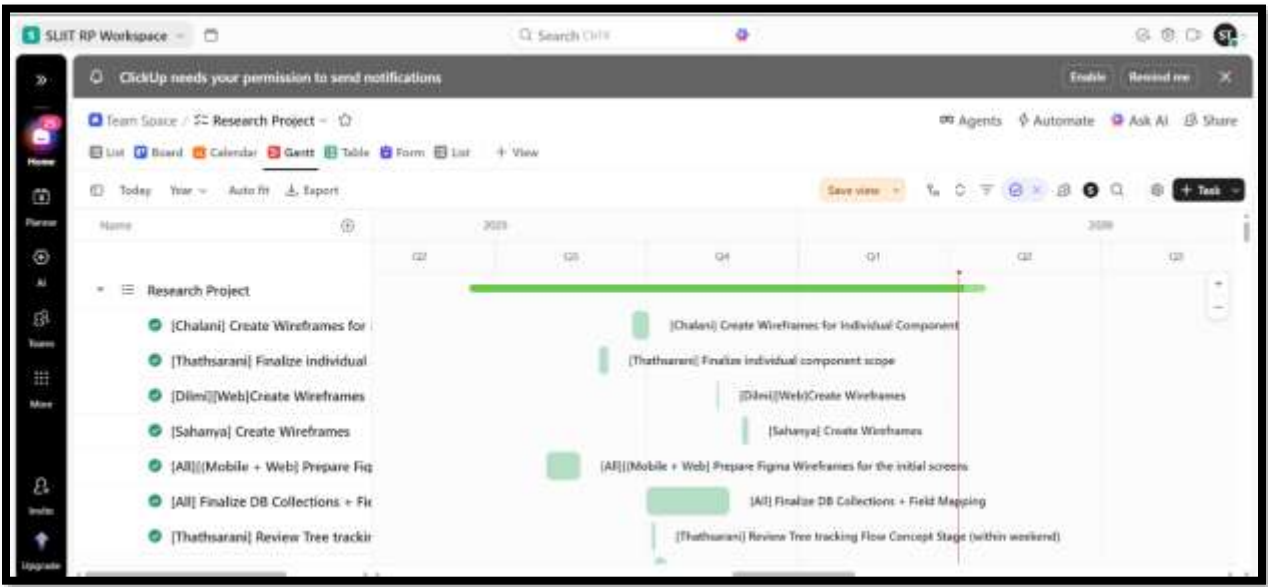


Figure 3.17: Clickup Milestone Tracking Table

3.7 Priority Distribution Table

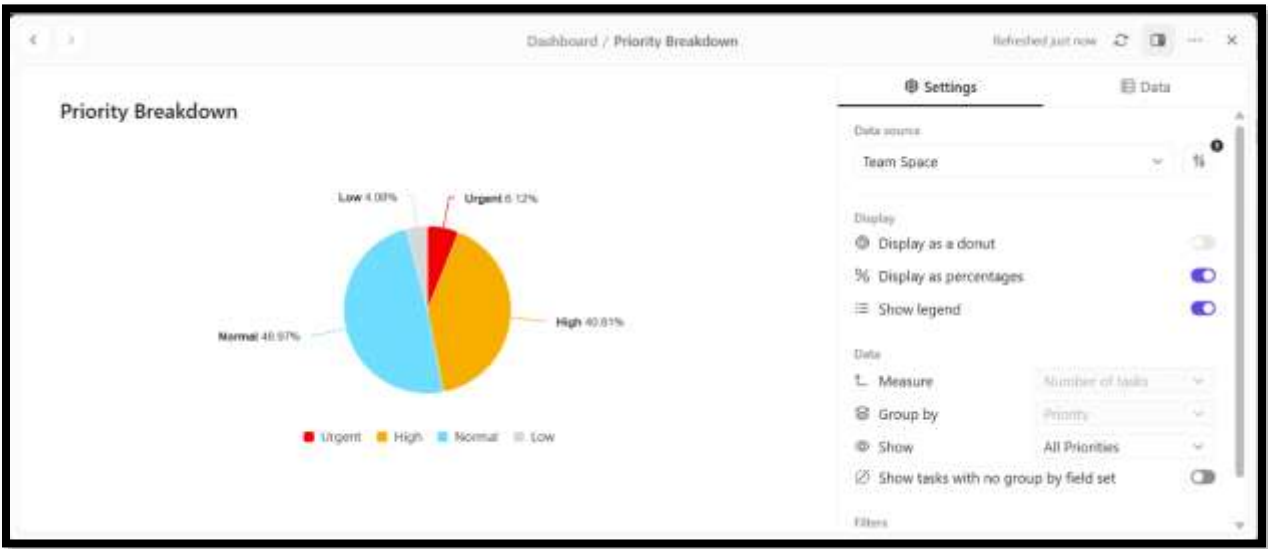


Figure 3.18: Clickup Priority Distribution Table

4. Resource & Work Management

4.1 Team Structure & Roles

A dedicated team managed Oudra's development, ensuring that all areas of the Decision Support System, from digital tree tracking, AI, and IoT to Blockchain and mobile development, were efficiently controlled. The roles were specified as the following:

- **WIJESINGHE A G T:**
Project Manager and overall leader
Responsible for digital tree tracking and virtual mapping.
- **DEHIGOLLA D N:**
IoT Integration Lead
Responsible for developing environmental sensor networks and growth factor analysis system.
- **AMARASOORIYA R P C A:**
AI/ML engineer
Responsible for developing, training, and deploying machine learning models for image-based resin assessment of trees.
- **THANNAKOON T A C S:**
Blockchain Lead
Responsible for the blockchain-based certification and traceability module.

All team members were in charge of the overall system architecture, project management of their respective components, and ensuring flawless integration across all modules. Quality assurance (QA) was dispersed throughout the team, with each member fully responsible for testing and confirming the reliability of their components prior to final system integration.

4.2 Work Distribution

Tasks were assigned in ClickUp using the distinct main components outlined in our system design. This ensured a balanced workload, unambiguous accountability, and concurrent progress.

- **Digital Tree Tracking & Virtual Mapping:** The mapping interfaces, tree tracking logic, and component-level testing to ensure precise geographic data handling.
- **IoT Sensor Network & Growth Analytics:** Managing the sensor data ingestion pipelines, real time analytics dashboards, and QA for all IoT data streams to ensure hardware, software synchronization.
- **AI-Assisted Resin Assessment:** Created image processing pipelines and trained AI models for resin detection with frameworks such as TensorFlow and Keras. To get high confidence scores, the model's accuracy was rigorously tested against numerous field image samples.
- **Blockchain Certification and Traceability:** Developed the smart contracts, certificate production, and verification interfaces, as well as rigorous testing of blockchain transaction latency and security.

4.3 Risk Register

Table 4.1: Risk Register Table

Risk ID	Risk Description	Probability	Impact
R01	Intermittent Network Connectivity: Losing internet access in remote plantation blocks.	High	High
R02	AI Accuracy Fluctuations: Inaccurate resin classification due to poor lighting in trunk photos.	Medium	High
R03	IoT Sensor Degradation: Environmental sensors drifting or failing over time in extreme outdoor weather.	Medium	Medium
R04	Blockchain Network Latency: Delays in generating certificates on the Polygon network.	Low	Medium
R05	Mobile Device Battery Drain: High battery consumption from continuous GPS tracking.	High	Medium
R06	Data Synchronization Conflicts: Overlapping offline edits from multiple workers causing data overwrites.	Medium	High
R07	Hardware Damage: NFC cards that are attached to trees being damaged by rain, animals, or fungal attacks.	High	Low
R08	Scope Creep: Project delays due to the complexity of integrating four highly distinct technologies.	Medium	High

4.4 Budget

Table 4.2: Budget in LKR

Resource Type	Description	Estimated Budget (LKR)	Actual Amount Spent (LKR)
Hardware & Devices	ESP32 components, IoT Sensors, NFC Tags, and testing devices	12,000	10,000
Field Visits	Travel and food expenses for data collection at Ayagama Pintanna Plantation	15,000	12,000
Miscellaneous	Internet data cards, backup storage, and unexpected minor expenses	10,000	9,000
Total		37,000	31,000

5. Communication Management

5.1 MS Teams Channel Structure

Microsoft Teams is the main communication and collaboration platform for the Oudra research project. The team works in a specialized MS Teams workspace called "Research Project" that was created using the SLIIT institutional account. The workspace is designed to provide both real-time collaboration and centralized file management via an integrated SharePoint/OneDrive document library accessible to all four team members and the research supervisors.

The document library is divided into three main layers: a shared General folder containing all common project materials, recording folder for storing all the online sync up meeting records and four individual member folders (Thathsarani, Dilmi, Sahanya, and Chalani), each of which includes component-specific materials and working files.

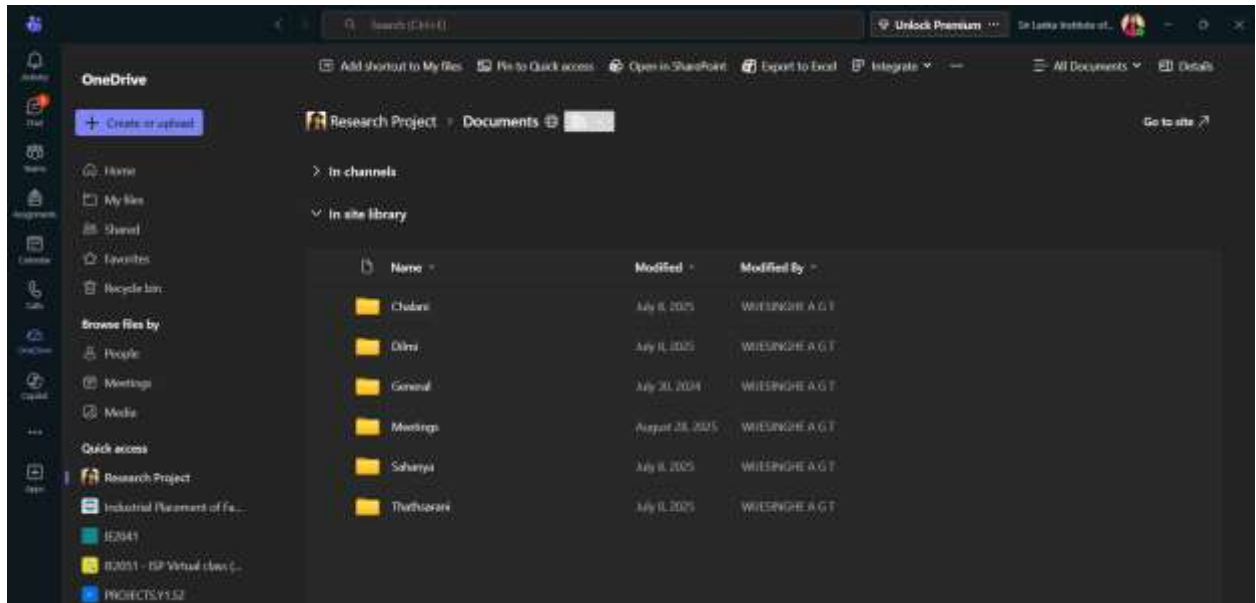


Figure 5.1: MS Teams Research Project Library Structure

5.1.1 Shared General Folder Structure

The General folder is the single source of truth for all shared project materials. It is divided into the following sub-folders, each having a distinct project management or research purpose:

Table 5.1: MS Teams General folder content

Folder	Content
1. Research Guidelines	SLIIT research project guidelines, marking rubrics, and submission requirements
2. Research Diary Log	Ongoing log of research activities, decisions, and progress maintained by the team
3. Potential Research Topic Ideas	Early-stage brainstorming documents from the project initiation phase
4. TAF	Topic Approval Form and related correspondence submitted for project topic approval
5. Letters	Formal correspondence including supervisor letters, field visit permission requests, and institutional communications
6. Field Visits	Documentation, photographs, and notes from the plantation field visit conducted at Pintanna Plantations
7. Wireframes	Initial wireframe files for all system screens
8. Initial User Interfaces	Early UI/UX wireframe designs and screen mockups for the web dashboard and mobile application
9. Required data set from the Pintanna Plantations	Pintanna PlantationThe raw dataset collected from Pintanna Plantations used for the system development initiation
10. Project Charter	The formal project charter document and the related materials that were approved at the start of the project
11. Oudra Logo	Official project logo assets in various formats used across presentations and documentation
12. Proposal Report	The written project proposal report and other related materials that were submitted to the research panel
13. Proposal Presentation	Slides and supporting materials from the initial project proposal presentation

14. Specialization Specific Requirements	ISE Specialization specific submission materials and documentations
15. Research Journal	The review paper materials that were submitted to the Elsevier (Science Direct) journals
16. Conference Papers	IEEE conference paper drafts, revision history, and submission records for TENSYP 2026, ICECET 2026...etc
17. Oudra Latest Codes	Shared code snapshots and reference implementations exchanged between team members during development
18. PP1	All materials related to Progress Presentation 1 - Presentation slides, Demo videos, Presentation Speech scripts, and feedback records
19. PP2	All materials related to Progress Presentation 2 - Presentation slides, Demo videos, Presentation Speech scripts, and feedback records
20. Recordings	Meeting recordings and session recordings from MS Teams calls

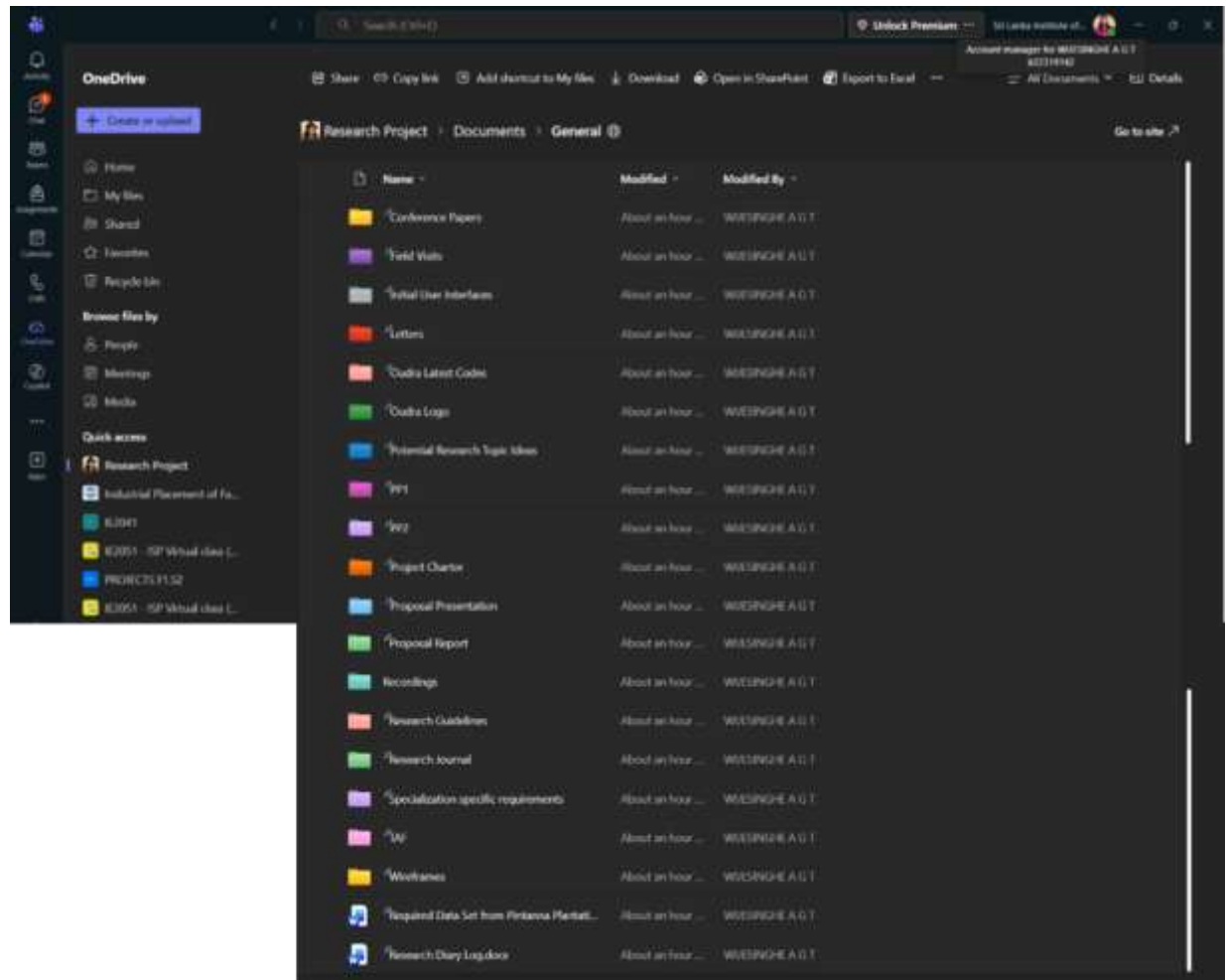


Figure 5.2: MS Teams – General Folder Structure

5.1.2 Individual Member Folders

Each team member has a personal folder within the common document library where they save component-specific working documents, draft reports, individual research notes, and development artifacts. This arrangement ensures that each member has their own workspace, yet the team has a complete view of all project materials in one centralized location.

Table 5.2: MS Teams – Individual Member Folder Content

Folder	Owner	Purpose
Thathsarani	WIJESINGHE A.G.T.	Component 1 working documents - GPS tracking, NFC integration, mobile app development artefacts
Dilmi	DEHIGOLLA D.N.	Component 2 working documents - IoT sensor data, ESP32 configurations, MQTT pipeline notes
Chalani	AMARASOORIYA R.P.C.A.	Component 3 working documents - AI model training logs, dataset preprocessing, resin detection notes
Sahanya	THENNAKOON T.A.C.S.	Component 4 working documents - smart contract drafts, blockchain deployment records, traceability dashboard notes

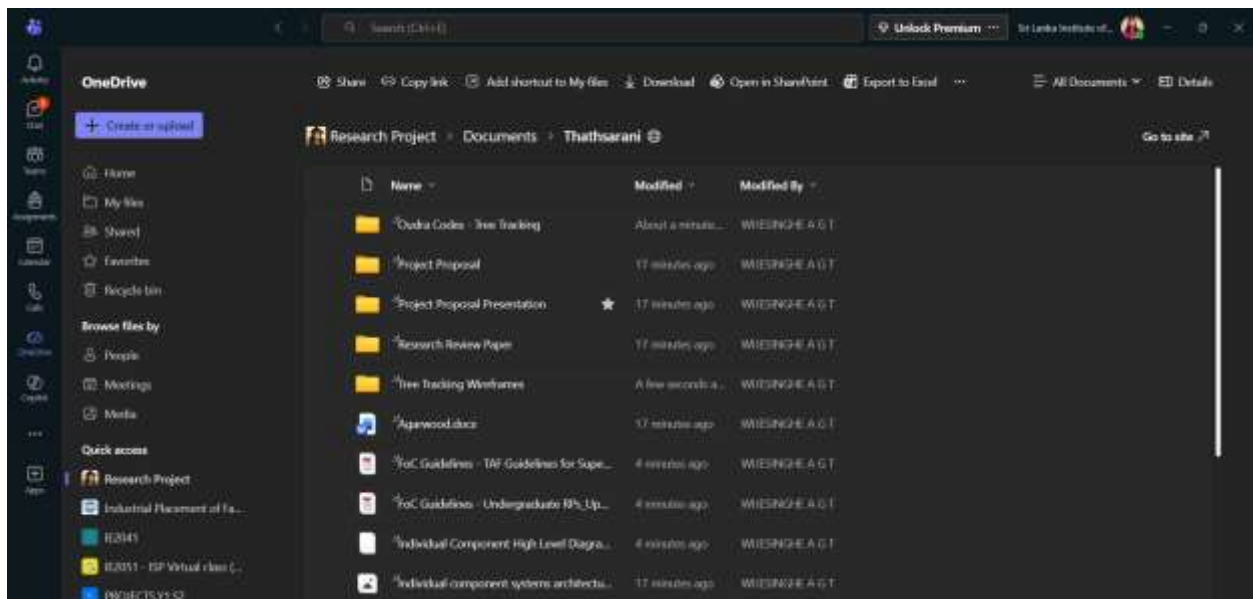


Figure 5.3: MS Teams – Sample Individual Member Folder Structure

5.1.3 Meetings Folder

A dedicated Meetings folder is maintained at the top level of the document library, separate from the General folder, to store all the online meeting recordings had between the team members and the supervisors, meeting minutes, agenda documents, and action item records from sprint planning sessions, supervisor progress reviews, and team retrospectives.

5.2 Meeting Log

The Oudra team kept a consistent meeting schedule throughout the project's lifecycle, with three types of meetings: supervisor meetings with Ms. Narmada Gamage and Ms. Chathurangika Kahandawaarachchi, external stakeholder meetings with domain experts and plantation management, and internal team meetings (sprint planning, conflict resolution, and integration sessions). All online meeting minutes are saved in the Meetings folder of the Research Project MS Teams document library.

The table below provides a chronological account of all meetings held from project inception to the current reporting period.

Table 5.3: Meeting Logs

Date	Meeting Type	Attendees	Key Decisions / Action Items
26/05/2025	Supervisor Meeting	All Members + Ms. Narmada Gamage + Ms. Chathurangika Kahandawaarachchi	Research topic finalized as "Smart Decision Support System for Sri Lankan Agarwood Plantations"; project scope and team component allocation discussed
19/06/2025	External Expert Meeting	All Members + Prof. Upul Subasinghe (Dean, Faculty of Applied Sciences, University of Sri Jayewardenepura)	Gained domain knowledge on agarwood cultivation, commercial value, and plantation challenges in Sri Lanka; research scope refined based on expert guidance
20/06/2025	Stakeholder meeting	All Members + Sadaharitha Plantations	Requested Sadaharitha Plantations to serve as resource person/client; declined to participate after discussions
21/06/2025	Stakeholder meeting	All Members + Pintanna Plantations	Requested Pintanna Plantation Pvt Ltd to serve as resource person/client; agreement initiated
25/06/2025	Stakeholder meeting	All Members + Manager - Ayagama	Formal research partnership confirmed with Pintanna Plantations;

		Estate Pintanna Plantations	first field visit arranged for 5th July 2025
08/07/2025	Supervisor Meeting	All Members + Supervisor	Findings from the first field visit presented; supervisor provided feedback on system requirements and research direction
14/08/2025	Supervisor Meeting	All Members + Supervisor	Project progress update presented; supervisor advised team to participate in the Value Proposition Workshop at CodeFest Technoprenuer Competition to deepen understanding of research commercialization
24/08/2025	Co-Supervisor Meeting	All Members + Co-Supervisor	Clarified doubts regarding the commercialization plan and project proposal draft document
25/08/2025	Supervisor Meeting	All Members + Supervisor	Project progress update; supervisor advised to finalize project proposal with suggested amendments, conduct a thorough literature review, and prepare the commercialization plan
15/09/2025	Internal Team Meeting	All Members	Decided to first prepare and submit a review paper to an Elsevier journal, followed by a research paper for IEEE conference submission
21/09/2025	Internal Team Meeting	All Members	Agreed on concurrent development of web app and mobile app with a common Node.js/Express backend and separate frontends; MongoDB Atlas selected as the database
27/09/2025	Supervisor Meeting	All Members + Supervisor	Project progress discussed; supervisor advised to set up GitHub repositories for backend and frontends, refine web interfaces based on her feedback, and begin keyword research and writing of the research journal review paper
09/10/2025	Supervisor Meeting	All Members + Supervisor	General project progress update presented to supervisor

10/10/2025	Stakeholder Meeting	All Members + Pintanna Plantations Manager	Discussed requirements for the second field visit; arrangements initiated for data acquisition visit
15/10/2025	Supervisor Meeting	All Members + Supervisor	Project progress update; supervisor advised team to prepare user interfaces for individual component-related functionalities
25/10/2025	Supervisor Meeting	All Members + Supervisor	Progress update; supervisor advised to continue literature review, begin writing the journal paper, and progress individual component system implementation
10/11/2025	Stakeholder Meeting	All Members + Pintanna Plantations Manager	Second field visit arranged for 15th November 2025
14/11/2025	Supervisor Meeting	All Members + Supervisor	Advised to gather required dataset during the upcoming Ayagama Plantation field visit, implement individual components based on collected data, prepare a working prototype for the next meeting, and continue writing the research review paper
21/11/2025	Internal Team Meeting	All Members	Planned next phases of system implementation across all four components
27/11/2025	Internal Team Meeting	All Members	Planned next implementation phases; resolved inter-branch merge conflicts; fixed API errors across components
03/12/2025	Internal Team Meeting	All Members	Planned next implementation phases; resolved branch conflicts and API errors; successfully merged multiple member branches into the main branch
09/12/2025	Internal Team Meeting	All Members	Planned next implementation phases; resolved branch conflicts and API errors

20/12/2025	Internal Team Meeting	All Members		Planned next implementation phases; resolved branch conflicts and fixed integration errors
24/12/2025	Internal Team Meeting	All Members		Planned next implementation phases; resolved branch conflict errors
29/12/2025	Internal Team Meeting	All Members		Planned next implementation phases; resolved API errors; successfully merged member branches into main branch
07/01/2026	Supervisor Meeting	All Members + Supervisor		Clarified doubts regarding research paper writing structure and content
12/01/2026	Internal Team Meeting	All Members		Planned next implementation phases; resolved API errors; successfully merged member branches into main branch
14/01/2026	Supervisor Meeting	All Members + Supervisor		General project progress update presented to supervisor
19/01/2026	Supervisor Meeting	All Members + Supervisor		Project progress update and discussion on journal paper progress and next steps
24/01/2026	Supervisor Meeting	All Members + Supervisor		Progress update; supervisor advised team to begin writing individual journal articles per component, continue system implementation, and complete system integration
27/01/2026	Internal Team Meeting	All Members		Planned next implementation phases; resolved API errors; successfully merged member branches into main branch
31/01/2026	Internal Team Meeting	All Members		Planned next implementation phases; resolved API errors; successfully merged member branches into main branch
06/02/2026	Internal Team Meeting	All Members		Planned next implementation phases; resolved API errors; merged

			branches; commenced SRS document preparation
10/02/2026	Internal Team Meeting	All Members	Planned next implementation phases; resolved API errors; successfully merged member branches
14/02/2026	Supervisor Meeting	All Members + Supervisor	Progress update; supervisor advised to prepare conference paper for ICECET 2026 and complete integration of individual component functionalities across the full application
20/02/2026	Internal Team Meeting	All Members	Planned next implementation phases; resolved API errors; merged branches successfully
23/02/2026	Internal Team Meeting	All Members	Planned next implementation phases; resolved API errors; merged branches successfully
28/02/2026	Internal Team Meeting	All Members	Planned next implementation phases; resolved API errors; merged branches successfully
04/03/2026	Internal Team Meeting	All Members	Planned next implementation phases; resolved API errors; merged branches successfully
07/03/2026	Supervisor Meeting	All Members + Supervisor	Progress update; supervisor asked team to prepare the final product for Progress Presentation 2 (PP2)
14/03/2026	Supervisor Meeting	All Members + Supervisor	Post-PP2 review meeting; supervisor advised to address all panel comments from PP2; directed team to conduct User Acceptance Testing (UAT) at Ayagama Pintanna Plantation and prepare the UAT report

6. Progress Summary

6.1 Achievements

Several important milestones across all system components have been achieved by the project team during the current development phase. The key achievements are as follows:

- Successfully designed and finalized the overall system architecture, providing seamless integration of IoT devices, AI models, blockchain modules, and web/mobile applications.
- Completed the System requirements Specification (SRS) and research proposal, defining system functionalities, user roles, and technical requirements.
- Using ESP32, created a functioning IoT-based sensor network device that can capture environmental factors such as temperature, humidity, soil wetness, and soil pH.
- Weather forecasting capabilities was integrated using external APIs, allowing the system to include future environmental conditions.
- Implemented the Digital Tree Tracking and Virtual Mapping System using NFC-based identification and GPS location tracking for accurate tree monitoring.
- Developed the initial version of the AI-based resin detection model, capable of classifying resin presence using tree trunk images.
- Initiated the Blockchain-Based Certification module, focusing on secure and tamper-proof record management for agarwood products.
- Established version control and collaboration workflows using GitHub, enabling efficient code management and team collaboration.
- Designed and developed initial web and mobile application interfaces to support user interaction for both plantation managers and field workers.
- Structured and managed project tasks effectively using ClickUp, ensuring proper task allocation, tracking, and progress monitoring.
- Improved the growth factor analysis logic, enabling the system to evaluate environmental conditions and support decision-making.
- Incorporated supervisor feedback from previous presentations (PP1) to refine component definitions and improve system clarity.

Overall, the achievements completed during this period demonstrate significant progress in both the technical implementation and research development of the project. The successful development of core components, along with the integration of IoT, AI, and supporting technologies, indicates that the project is progressing in alignment with the planned objectives and milestones. These accomplishments provide a strong foundation for the next phase, which will focus on full system integration, performance enhancement, and final validation.

6.2 Challenges Faced & Resolutions Table

The team encountered several technical and coordination challenges during the development process. However, effective measures were established to address these difficulties and ensure ongoing project progress.

Table 6.1: Challenges faced & Resolutions

Challenge	Description	Impact	Resolution
Sensor Data Accuracy	Inconsistent or fluctuating sensor readings during initial testing	Reduced reliability of environmental analysis	Sensors were calibrated and tested under different conditions to improve accuracy
Hardware Integration Issues	Difficulty connecting multiple sensors with ESP32	Delayed IoT module implementation	Optimized circuit design and used compatible libraries for stable communication
Weather API Integration	Intermittent failures in retrieving real-time weather data	Affected analysis accuracy	Implemented error handling and fallback mechanisms for data retrieval
Component Understanding	Initial confusion regarding scope and functionality of components	Affected presentation clarity	Revised component definitions based on supervisor feedback
AI Model Performance	Low prediction accuracy in early stages	Reduced effectiveness of resin detection	Improved dataset quality and retrained the model
Team Coordination	Managing multiple components simultaneously	Caused delays in task completion	Improved planning using ClickUp and regular team discussions
Network Connectivity Issues	Unstable internet connection in plantation environments	Affected real-time data transmission	Implemented data caching and synchronization mechanisms

Data Synchronization	Delays in syncing mobile app data with backend	Caused inconsistency in displayed data	Implemented periodic sync and conflict resolution strategies
Limited Dataset Availability	Insufficient data for training AI models	Reduced model accuracy	Expanded dataset using additional samples and data augmentation
Integration Complexity	Difficulty integrating IoT, AI, and blockchain components	Slowed overall system development	Followed modular development approach and tested components independently
UI/UX Design Challenges	Difficulty designing user-friendly interfaces for both workers and managers	Reduced usability	Conducted iterative design improvements based on feedback
Time Management Constraints	Balancing multiple academic tasks with project work	Delayed some deliverables	Used ClickUp scheduling and prioritization to manage workload effectively
Testing Environment Limitations	Lack of real plantation environment for testing	Limited real-world validation	Simulated environmental conditions and conducted controlled testing
API Rate Limitations	Restrictions on number of API calls for weather data	Interrupted continuous data retrieval	Optimized API usage and implemented caching mechanisms
Error Handling in System	Unexpected system errors during integration	Reduced system stability	Implemented proper exception handling and logging mechanisms

6.3 Upcoming Tasks – Next Sprint

The system's core functionalities have been successfully developed; thus, the next sprint will focus on system refinement, validation, and final preparation for evaluation. The key tasks scheduled for the upcoming phase are as follows:

- Improve the accuracy and reliability of the IoT sensor data processing and growth factor analysis module through further calibration and testing.
- Enhance the performance and prediction accuracy of the AI-based models by refining datasets and optimizing model parameters.
- Conduct User Acceptance Testing (UAT) in a real plantation environment to validate system functionality and usability under practical conditions.
- Collect feedback from plantation users and stakeholders to evaluate system usability, effectiveness, and practical applicability.
- Identify and resolve any system errors or performance issues identified during testing to ensure system stability.
- Optimize the overall system integration, ensuring smooth data flow between IoT devices, backend services, and user interfaces.
- Finalize and improve web and mobile application interfaces based on user feedback and usability considerations.
- Complete all final documentation, including technical reports, user manuals, and project deliverables.
- Prepare final presentation materials, including slides, demonstrations, and supporting documents for evaluation.

This phase will ensure that the system is fully optimized, validated, and ready for final deployment and assessment, aligning with the project's overall objectives.

7. Research Panel Feedback

7.1 PP1 Feedback

Table 7.1: Progress Presentation 1 Feedback

Comments	Actions / Solutions
Link Tree ID with block	Implemented new Tree ID format from T-000001 to TA-000001 block is now embedded directly in every tree identifier
More data needed for analysis	Expanded dataset with field-collected bark imagery; IoT nodes now capturing continuous historical sensor readings for trend analysis and factor correlation
Pay more attention to testing	Conducted unit testing per module, integration testing across API endpoints, and AI model validated at 95% accuracy on test dataset
Factor analysis needed and Confusion about component explanation	The component explanation was improved by introducing growth factor analysis and clearly defining the system functionality and outputs

7.2 PP2 Feedback

Table 7.2: Progress Presentation 2 Feedback

Comments/Feedback	Actions / Solutions
During the UAT, capture the actual GPS coordinates of trees and display the plantation map	Successfully captured real-time GPS coordinates for each tree during UAT and integrated them into the system to visualize locations on a plantation map
During the UAT, test the IoT device in the plantation, capture sensor values, and compare them with actual environmental values	Conducted on-field IoT testing, collected live sensor data (temperature, humidity, soil moisture), and validated accuracy by comparing with actual environmental measurements

Train the model again and include a display mechanism to identify invalid images	Retrained the AI model with an improved dataset and implemented a validation layer to detect and flag invalid or low-quality images
Obtain a confirmation letter to verify that the UAT has been successfully completed	Collected official confirmation and approval from the plantation authority to verify that UAT was successfully completed

8. Lessons Learned

8.1 Successes

- **Effective Agile Management via ClickUp:** Using ClickUp to enforce stringent sprint planning and workload division was extremely successful. It enabled four independent, complicated technical domains (AI, IoT, Blockchain, and GIS Mapping) to be developed concurrently without incurring significant scheduling obstacles.
- **Seamless Multidisciplinary Integration:** Early on in the project, the team established a unified system architecture and central MongoDB Atlas schema, allowing them to successfully combine widely different technologies into a single coherent Node.js/Express backend without catastrophic structural breakdowns.
- **Version Control Discipline:** Strict adherence to the GitFlow branching method on GitHub prevented catastrophic code overwrites. Creating separate feature branches for each component kept individual development segregated and secure until integration testing.
- **Real-World Technical Applicability:** The successful implementation of an offline-first architecture with background synchronisation in the React Native mobile app demonstrated great efficacy, directly addressing the real-world constraint of inadequate internet connectivity in rural agarwood plantations.

8.2 Challenges

- **Integration Merge Conflicts:** While individual feature branches were well managed, merging all four large components into the central dev branch during Sprint 8 caused complex API payload mismatches and dependency conflicts, necessitating unexpected, intensive collaborative debugging sessions.
- **Environmental Hardware Calibration:** Adapting the physical ESP32 microcontrollers and IoT sensors to endure simulated Sri Lankan plantation weather conditions (high humidity, heavy rain) resulted in unforeseen delays and necessitated numerous hardware changes and protective casing modification.

- **Steep Domain Learning Curves:** Integrating advanced deep-tech concepts, specifically deploying smart contracts on the Polygon Layer-2 network and optimizing machine learning pipelines, presented a steeper learning curve than expected, necessitating significantly more research hours early on in the project.
- **API Quotas and Financial Constraints:** During the testing of the Digital Tree Tracking module, frequent real-time route optimization requests quickly exceeded free-tier Google Maps API quotas, forcing the team to use aggressive caching tactics and dummy data to stay inside budget.
- **Data Synchronization Overwrites:** Handling offline data updates from numerous field workers at the same time caused early data overwrites in the database, forcing the team to suspend UI development and totally restructure the backend's timestamp-based conflict resolution mechanism.

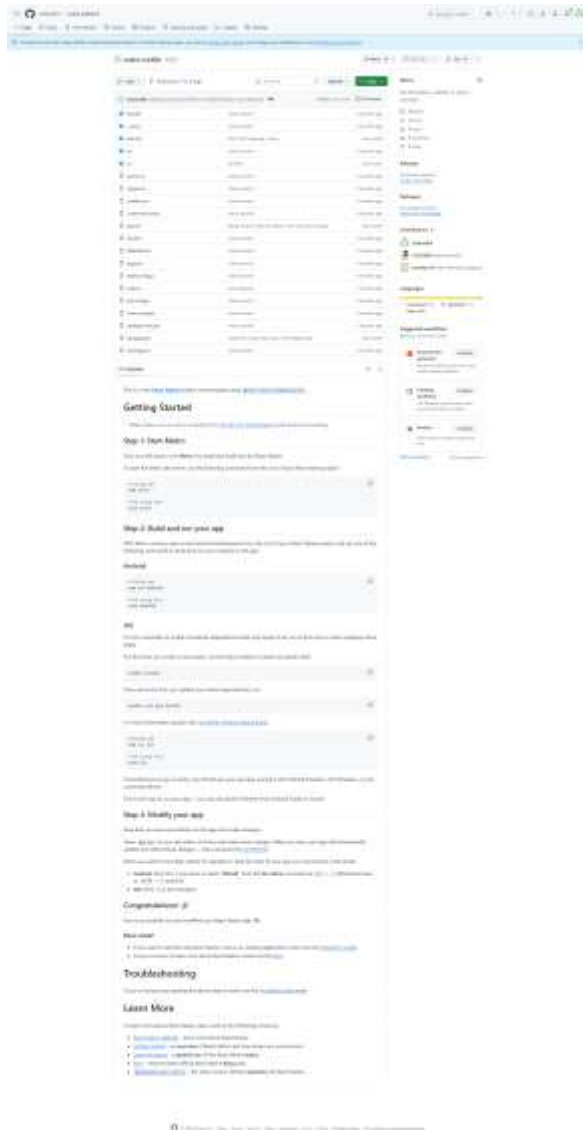
8.3 Improvements for Future Projects

- **Implementation of CI/CD Pipelines:** In future iterations, automating component testing would be achieved by building up Continuous Integration and Continuous Deployment (CI/CD) pipelines via GitHub Actions earlier in the project. This would detect cross-module API integration errors immediately after code commit, saving significant manual QA work.
- **Advanced PM Tool Automations:** While ClickUp worked well for task tracking, future projects should take advantage of its advanced automation features (for example, automatically updating a task to "In Review" when a GitHub Pull Request is opened) to reduce the manual administrative overhead of updating the backlog.
- **Earlier User Acceptance Testing (UAT):** Bringing in actual end-users (or replicating field worker limitations) earlier in the prototyping phase would assist optimize the mobile application's ergonomics, notably optimizing the UI for outdoor, high-glare situations before finalizing the code.

APPENDICES

Appendix A – GitHub Branch Summary

1. Oudra – Mobile Application



2. Oudra – Web Application

The screenshot shows the GitHub repository page for 'oudra_client' by user 'Chaitan010'. The repository has 12 branches and 4 tags. The main branch is 'main'. The repository is a JavaScript project, as indicated by the 'JS Code' button. The repository description is 'Oudra Client'. The repository is public. The repository is created on April 24, 2021. The repository is a JavaScript project, as indicated by the 'JS Code' button. The repository is public. The repository is created on April 24, 2021.

Repository Details:

- Repository: oudra_client
- Owner: Chaitan010
- Created: April 24, 2021
- Updated: April 24, 2021
- Size: 1.5 MB
- Language: JavaScript

Files and Folders:

- github/workflows: Add REACT_APP_API_URL to build workflow (last week)
- public: Build static assets (last week)
- src: Update frontendClientManager.js (last week)
- api: Initial commit (7 months ago)
- apiService: Integrated the Sample Map API (4 months ago)
- README.md: Initial commit (7 months ago)
- package-lock.json: Initial commit (7 months ago)
- package.json: Initial commit (7 months ago)
- src/index.js: Initial commit (7 months ago)
- src/index.css: Initial commit (7 months ago)

README:

Getting Started with Create React App

This project was bootstrapped with [Create React App](https://github.com/facebook/create-react-app). Available Scripts In the project directory, you can run: `npm start` Runs the app in the development mode. Open http://localhost:3000 to view it in your browser. The page will reload when you make changes. You may also see any lint errors in the console. `npm test` Launches the test runner in the interactive watch mode. See the section about [running tests](https://facebook.github.io/create-react-app/docs/running-tests) for more information. `npm run build` Builds the app for production to the `build` folder. It correctly bundles React in production mode and optimizes the build for the best performance. The build is minified and the filenames include the hashes. Your app is ready to be deployed! See the section about [deployment](https://facebook.github.io/create-react-app/docs/deployment) for more information. `npm run eject` Note: this is a one-way operation. Once you `eject`, you can't go back! If you aren't satisfied with the build tool and configuration choices, you can `eject` at any time. This command will remove the single build dependency from your project. Instead, it will copy all the configuration files and the transitive dependencies (webpack, Babel, ESLint, etc) right into your project so you have full control over them. All of the commands except `npm run eject` will still work, but they will point to the copied scripts so you can tweak them. At this point you're on your own. You don't have to ever use `npm run eject`. The scattered features are suitable for small and middle deployments, and you shouldn't feel obligated to use this feature. However we understand that this tool wouldn't be useful if you couldn't customize it when you are ready for it. ## Learn More You can learn more in the [Create React App documentation](https://facebook.github.io/create-react-app/docs/getting-started). To learn React, check out the [React documentation](https://reactjs.org/docs/getting-started.html). ## Code Splitting This section has moved here: https://facebook.github.io/create-react-app/docs/code-splitting. ## Analyzing the Bundle Size This section has moved here: https://facebook.github.io/create-react-app/docs/analyzing-the-bundle-size. ## Making a Progressive Web App This section has moved here: https://facebook.github.io/create-react-app/docs/making-a-progressive-web-app. ## Advanced Configuration This section has moved here: https://facebook.github.io/create-react-app/docs/advanced-configuration. ## Deployment This section has moved here: https://facebook.github.io/create-react-app/docs/deployment. ## npm run build fails to minify This section has moved here: https://facebook.github.io/create-react-app/docs/troubleshooting-npm-run-build-fails-to-minify. # Oudra client

3. Oudra – Server

The screenshot displays the GitHub repository for **Oudra/Oudra-server**. The repository has 15 watchers and 8 tabs. The main content area shows a list of files and folders, including `ghrhub/wordfence`, `app`, `model`, `uploads`, `ghrhub`, `README.md`, `node_modules`, `package-lock.json`, `package.json`, `resources/headers`, `resources/js`, `utils/headers`, `utils/js`, `utils/headers`, and `utils/js`. The `README.md` file is selected, showing the **Glimmer-server** logo and text.

Files and Folders:

- `ghrhub/wordfence`: Add or update the Wordfence Service Audit and Configuration...
- `app`: KIT Run data
- `model`: Run detection
- `uploads`: Initial content
- `ghrhub`: Initial content
- `README.md`: Initial content
- `node_modules`: Install server certificate and blockchain verification complete...
- `package-lock.json`: Initial content
- `package.json`: Development for 2
- `resources/headers`: Install server certificate and blockchain verification complete...
- `resources/js`: Install server certificate and blockchain verification complete...
- `utils/headers`: Install server certificate and blockchain verification complete...
- `utils/js`: Run JS
- `utils/headers`: Install server certificate and blockchain verification complete...
- `utils/js`: Install server certificate and blockchain verification complete...

README:

Glimmer-server

About:

Not a description, website, or logo provided.

Releases:

No releases published. [Create a new release](#)

Packages:

No packages published. [Publish your first package](#)

Contributors: 4

- Oudra/O
- R22B542 Wjwongh42
- narvaya-01 Odris Nwanya Gargale
- Clahanga Gargale Nwanya

Languages:

JavaScript 70.1% Python 2.9%

4. Oudra – AI Server

